Rakshak S Barkur

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Education

BMS Institute of Technology & Management, Bengaluru Sep 2023 - May 2027
BE in Department of Artificial Intelligence and Machine Learning CGPA - 8.5
Relevant Coursework: DSA, AI, DBMS, OS, OOPS with Java, CN, MLOps, MongoDB & Git.

Technical Strengths

Languages : Python, Java, Kotlin, C/C++, HTML, CSS, & JavaScript
Databases : SQL, MongoDB, Qdrant, FAISS, PostgreSQL & Redis
Libraries : STL (C++), LlamaIndex, Hugging Face Transformers, Flower, Numpy, Scikit-Learn, TensorFlow, Pandas & Matplotlib
Certifications : C++, Python, DSA, Google AI Essential, Google Prompt Engineering, CP & MySQL
Tools : VSCode, LaTeX, Excel, Jupyter Notebook & MATLAB

Experience

Circles - Data Engineer, Intern Aug'26 - Present

- Architected and deployed scalable Python-based ETL pipelines with Apache Airflow, Spark, Iceberg and AWS S3, streamlining large-scale data processing and optimizing batch workflows for improved performance and reliability.
- Engineered robust API-driven integration workflows to ingest complex XML/SOAP data feeds, implementing record-level parsing and comprehensive exception handling to ensure reliable data ingestion and prevent data loss.
- Containerized and managing cloud-native data workflows on Kubernetes, integrating automated data validation frameworks to boost pipeline reliability and maintain data integrity across production environments.

TestAing - AI/ML, Intern Mar'26 - Jul'26

- Architected a multi-agent AI system using LangGraph and FastAPI, implementing a supervisor node with dynamic routing, parallel execution across 4 specialized agents and confidence-based decision policies
- Engineered a low-latency agent-to-frontend state synchronization pipeline using Server-Sent Events (SSE), CopilotKit and RFC 6902 JSON Patch, enabling seamless rendering of schema-driven GenUI/A2UI and controlled React components while cutting network payload overhead

Conatus Bharat - Full Stack Developer, Intern Sep'25 - Mar'26

- Architected and engineered a full-stack education platform using a modular TypeScript, NestJS, Next.js & PostgreSQL architecture; Implemented role-based REST APIs with JWT/bcrypt authentication to ensure secure, low-overhead session handling.
- Spearheaded backend and database performance tuning through query profiling, strategic indexing & schema refactoring, slashing average API latency by 88% (from 25ms to 3ms) & significantly expanding system throughput under peak load.

Projects

Agentic-ICU: Multi-Agent ICU Deterioration Monitoring /Agentic-ICU

- **Engineered predictive deep learning and ML pipelines** on the PhysioNet 2019 dataset to forecast early sepsis with an AUROC of 0.923 and respiratory failure with an AUROC of 0.961. Counteracted severe ∼1% class imbalance via SMOTE and weighted sampling, extracting 292 temporal features with integrated SHAP explainability.
- **Architected a 5-agent clinical decision framework** incorporating a 9-check rule-based Signal Quality artifact gate, vector search via Qdrant & a logistic-regression ensemble of calibrated model outputs (**0.931 validation AUROC**) to eliminate false positives.
- **Implemented an automated Clinical Reasoner** agent that synthesizes multi-modal risk scores into auditable `CRITICAL` / `WATCH` / `STABLE` alerts, generating plain-English clinical rationales and suggested triage interventions for rapid physician response.
- **Developed a real time full stack streaming pipeline** using **FastAPI and WebSockets** to power an interactive dashboard with risk timelines, SHAP & saliency maps. Secured endpoints via rate limiting & API keys, ensuring production stability with **210 automated pytest tests**.

Early Detection of Neonatal Sepsis using Federated Learning /Neonatal-Sepsis Accepted into ICICI'26

- Implemented a privacy-preserving federated learning framework for early neonatal sepsis prediction, enabling secure multi-hospital EHR training using Flower and FedAvg, ensuring HIPAA- and GDPR-compliant collaboration.
- Developed GRU-D and Transformer time-series models on 40 NICU clinical features, improving AUROC from 0.829 → 0.893 and accuracy from 73.9% → 94.7% to enhance early detection reliability.
- Strengthened clinical decision support by increasing precision from 0.187 to 0.712 (∼280) and reducing the false-positive rate from 0.2422 to 0.0147, minimising alarm fatigue and unnecessary interventions.

RAG: Agentic Compliance RAG Engine /RAG

- Architected a Python Agentic RAG system with a multi-stage reasoning loop for query refinement, improving answer relevance for over 1,000 unique data sources.
- Engineered a high-precision retrieval engine using Qdrant, implementing Hybrid Search (Dense + BM25) with Reciprocal Rank Fusion (RRF) and FlashRank cross-encoder reranking for semantic and exact-keyword queries.
- Optimized performance and cost through a tiered LLM strategy, using local Mistral (Ollama) for reasoning and fact-checking while reserving Gemini 2.0 Flash for final synthesis, reducing cloud API consumption by ∼80%.

Folibuddy: AI-Powered Resume to Portfolio Generator /Folibuddy Live Demo `@folibuddy`

- Automated conversion of PDF resumes into deployable personal websites, implementing backend in Python, backed with FastAPI and Google Gemini API to orchestrate parsing and site generation.
- Implemented structured LLM outputs for resume parsing and designed an editable input interface enabling users to review & personalise extracted content before publishing the site to a custom domain/URL.

BMSIT Faculty App /BMSIT-Faculty-app

- Built a Faculty Onboarding System using FastAPI & React.js, implementing Role-Based Access Control (RBAC) to manage administrative and faculty workflows.
- Implemented authentication and security workflows using JWT for session management, bcrypt password hashing and a multi-step first-login flow with automated temporary credentials and expiry logic.
- Designed and documented RESTful APIs backed by MongoDB, integrating an SMTP email service with mock mode and delivering full CRUD functionality for user profile management.

Extra Curricular Activities

Competitive Programming:

- </> LeetCode : Solved 480+ problems, Peak Rating 1953.
- </> GeeksforGeeks : Solved 700+ problems

Positions of Responsibility: Serving as Chair & VP of Alterino, BMSIT's technical Club, leading event organisation and logistics.
Open Source Contributor (GSSoC): Contributed to open-source projects & ranked 65th globally among ∼50,000 contributors.